

Thème : Architecture réseau et segmentation d'une PME

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- 2) Topologie réseau (schéma)
- 3) Configuration technique
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I. Scénario réseau de la PME

1) Contexte

Une petite entreprise avec 3 départements principaux et un réseau invité pour les visiteurs. L'objectif est de **segmenter le réseau** pour améliorer la sécurité et l'organisation.

2) Départements et Vlan

Département	VLAN	Sous-réseau IP	Passerelle	Nombre de PC
IT	10	192.168.10.0/24	192.168.10.1	02
Comptabilité	20	192.168.20.0/24	192.168.20.1	02
RH	30	192.168.30.0/24	192.168.30.1	02
Wi-Fi invité	99	192.168.99.0/24	192.168.99.1	02

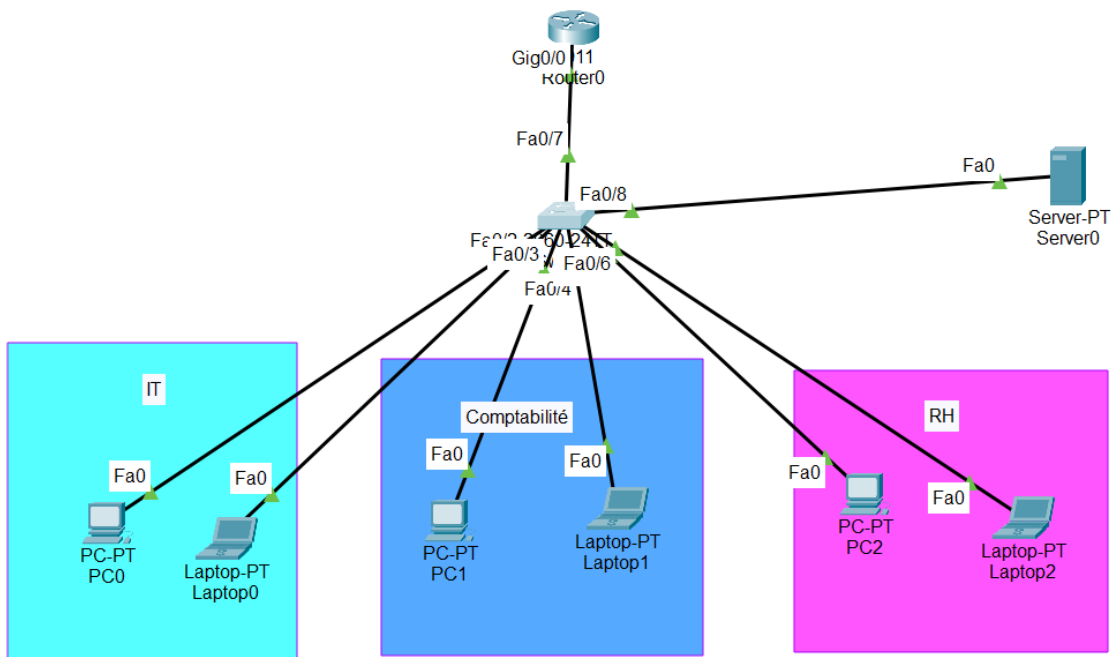
3) Matériel utilisé

- 1 switch manageable (pour VLANs)
- 1 routeur (inter-VLAN routing)
- 1 serveur DHCP
- 6 PC (2 par département)
- 1 point d'accès Wi-Fi pour le réseau invité (optionnel)

4) Objectifs pédagogiques du projet

- Créer des VLANs pour chaque département.
- Assigner les ports des PC aux VLANs correspondants.
- Configurer le routeur pour permettre la communication inter-VLAN.
- Installer un serveur DHCP pour attribuer les IP automatiquement les PC récupèrent automatiquement leur IP
- Tester la connectivité et vérifier la segmentation.

II. Schéma réseau



III. Configuration du switch VLAN

1) Création des VLANs

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Comptabilite
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name RH
Switch(config-vlan)#exit
Switch(config)#vlan 99
Switch(config-vlan)#name Wifi_Invite
Switch(config-vlan)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

2) Assignment des ports aux VLANs

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface range fa0/5-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#interface fa
Switch(config)#interface fa0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 99
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

NB :

- Chaque département est sur son VLAN.
- Les broadcasts sont isolés.

- Le Wi-Fi invité est séparé du réseau interne.

3) Mettre le serveur dans le bon vlan

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/8
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

4) Configuration du switch

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface Gig
Switch(config)#interface GigabitEthernet0/0
%Invalid interface type and number
Switch(config)#interface fa
Switch(config)#interface fastEthernet0/7
Switch(config-if)#switch
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

Switch(config-if)#switchport trunk allowed vlan 10,20,30,99
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#
```

5) Vérifier les VLANs

```
Switch>enable
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2
20	Comptabilite	active	Fa0/3, Fa0/4
30	RH	active	Fa0/5, Fa0/6
99	Wifi_Invite	active	Fa0/7
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

IV. Configuration du routeur pour l'inter-VLAN routing

1) Configuration de l'interface physique et les sous interfaces VLAN

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabit
Router(config)#interface gigabitEthernet0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255?
A.B.C.D
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet0/0.99
Router(config-subif)#encapsulation dot1Q 99
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
```

Explications :

- Chaque VLAN a sa sous-interface avec **encapsulation dot1Q**.

- L'adresse IP de la sous-interface est **la passerelle de ce VLAN**.

2) Activer l'interface principale

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.99, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
```

NB :

- Les VLANs peuvent communiquer via le routeur si nécessaire.
- La communication interne reste isolée selon le VLAN (segmentation efficace).

3) Utiliser ip helper-address sur le routeur

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0.10
Router(config-subif)#ip helper-address 192.168.10.2
Router(config-subif)#exit
Router(config)#interface g0/0.20
Router(config-subif)#ip helper-address 192.168.10.2
Router(config-subif)#exit
Router(config)#interface g0/0.30
Router(config-subif)#ip helper-address 192.168.10.2
Router(config-subif)#exit
Router(config)#interface g0/0.99
Router(config-subif)#ip helper-address 192.168.10.2
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Router#wr
Building configuration...
[OK]
Router#

```

4) Vérifier la configuration

```

Router>enable
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/0.99	192.168.99.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

Router#

```

V. Configuration du serveur DHCP

1) Configurer les pools DHCP pour chaque VLAN

On crée un pool pour chaque VLAN avec :

- Adresse réseau
- Masque
- Passerelle (IP du routeur pour le VLAN)
- Plage d'adresses disponibles

VLAN	Nom du pool	Réseau	Masque	Passerelle	Plage IP disponible
------	-------------	--------	--------	------------	---------------------

10	IT	192.168.10.0	255.255.255.0	192.168.10.1	192.168.10.10 → 192.168.10.50
20	Comptabilité	192.168.20.0	255.255.255.0	192.168.20.1	192.168.20.10 → 192.168.20.50
30	RH	192.168.30.0	255.255.255.0	192.168.30.1	192.168.30.10 → 192.168.30.50
99	Wi-Fi invité	192.168.99.0	255.255.255.0	192.168.99.1	192.168.99.10 → 192.168.99.50

2) Attribuer une adresse statique au serveur

Servero

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:63FF:FEE7:EB24

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

3) Configurer le DHCP sur le serveur

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Wi-Fi invité

Default Gateway: 192.168.99.1

DNS Server: 8.8.8.8

Start IP Address : 192 168 99 10

Subnet Mask: 255 255 255 0

Maximum Number of Users : 41

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Wi-Fi invité	192.168.99.1	8.8.8.8	192.168.9...	255.255.2...	41	0.0.0.0	0.0.0.0
RH	192.168.30.1	8.8.8.8	192.168.3...	255.255.2...	41	0.0.0.0	0.0.0.0
Comptabilité	192.168.20.1	8.8.8.8	192.168.2...	255.255.2...	41	0.0.0.0	0.0.0.0
IT	192.168.10.1	8.8.8.8	192.168.1...	255.255.2...	41	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	169.254.0.0	255.255.0.0	512	0.0.0.0	0.0.0.0

4) Configurer les PC pour DHCP

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

▼

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

192.168.30.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.30.1

DNS Server

8.8.8.8

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::20C:CFFF:FEE6:ABAE

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

▼

Username

Password

☐ Top

Physical Config Desktop Programming Attributes

IP Configuration X

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

192.168.20.11

Subnet Mask

255.255.255.0

Default Gateway

192.168.20.1

DNS Server

8.8.8.8

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

 /

Link Local Address

FE80::201:64FF:FE74:359C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

Physical Config Desktop Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.10.4

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::2E0:F9FF:FE65:A96B

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

VI. Tests de connectivité

1) Ping entre PC du même VLAN

```
C:\>ping 192.168.10.4
```

```
Pinging 192.168.10.4 with 32 bytes of data:
```

```
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.10.4:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>|
```

```
C:\>ping 192.168.20.11
```

```
Pinging 192.168.20.11 with 32 bytes of data:
```

```
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.20.11: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.20.11: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.20.11:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>|
```

```
C:\>ping 192.168.30.10
```

```
Pinging 192.168.30.10 with 32 bytes of data:
```

```
Reply from 192.168.30.10: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.30.10: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.30.10: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.30.10: bytes=32 time=18ms TTL=128
```

```
Ping statistics for 192.168.30.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 18ms, Average = 4ms
```

```
C:\>
```

2) Ping entre VLANs (inter-VLAN)

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

3) Ping vers le serveur

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Conclusion :

Ce projet a permis de créer un réseau segmenté pour une PME, avec des VLANs pour chaque département et un serveur DHCP pour attribuer automatiquement les adresses IP. Les tests de connectivité ont confirmé que le routage inter-VLAN fonctionne correctement et que le réseau invité peut rester isolé. Ce projet illustre l'importance de la segmentation pour la sécurité et l'organisation d'un réseau professionnel.

FIN DU PROJET